PyRMD Studio: User Guide

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1. Introduction

1.1. General Overview

PyRMD is a Ligand-Based Virtual Screening (LBVS) tool written in Python, powered by the Random Matrix Discriminant (RMD) machine learning algorithm. The project is developed by the Cosconati Lab at the University of Campania "Luigi Vanvitelli". This protocol is outlined in Section 5 of this guide.

A new protocol, **PyRMD2Dock**, has been devised to combine PyRMD with any docking software, enabling effective high-throughput virtual screening of ultra-large molecular databases. This protocol is outlined in Section 6 of this guide.

For a complete scientific explanation of PyRMD and PyRMD2Dock, please refer to our open-access manuscripts published in the *Journal of Chemical Information and Modeling*:

- **PyRMD:** https://pubs.acs.org/doi/full/10.1021/acs.jcim.1c00653
- **PyRMD2Dock:** https://pubs.acs.org/doi/full/10.1021/acs.jcim.3c00647

This guide mainly focuses on the new **PyRMD Studio** Graphical User Interface (GUI), a user-friendly tool designed for effective high-throughput virtual screening of ultra-large molecular databases in drug discovery campaigns.

1.2. Software Description

The PyRMD Studio GUI addresses the need for accessible tools in computational chemistry that can efficiently handle large-scale molecular databases. The graphical interface eliminates the need for extensive command-line expertise, making it accessible to a broader range of researchers.

The software architecture uses PyQt5 for the graphical interface and integrates with the RDKit cheminformatics library for molecular manipulation. Dependency management and installation are simplified through Conda.

Key Features:

- **Dual-Mode User Interface:** The software features a dual-mode interface to accommodate varying levels of user expertise.
  - **Beginner Mode:** Provides a streamlined interface with essential parameters, comprehensive guidance, and tooltips. This mode is designed to help new users successfully complete virtual screening projects while preventing common errors.
  - **Expert Mode:** Unlocks full parameter control and advanced features, including

fingerprint configuration, statistical analysis parameters, custom epsilon cutoff strategies, and batch processing capabilities.

- **Configuration Management:** The system uses .ini files to store all input parameters in an organized, hierarchical structure. This allows users to save, reuse, and reproduce specific parameter configurations across different projects.
- **Database Integration:** The GUI provides comprehensive database integration, supporting both public databases and custom user datasets.
  - **ChEMBL Parser:** An automated parser enables direct access to the ChEMBL database with customizable filtering and selection criteria based on target proteins, activity types, and data quality.
  - **Custom Dataset Support:** Users can import proprietary compound libraries or literature-derived datasets by providing them in a standard .csv file format.

2. System Requirements

This section outlines the necessary hardware and software for running PyRMD Studio GUI.

2.1. Operating System Requirements

- **Recommended OS:** Ubuntu 20.04 LTS or newer, Windows 11 or newer.
- **Desktop Environment:** A standard graphical desktop environment is required for the GUI. Headless server configurations are possible for batch processing but require additional setup.

2.2. Hardware Requirements

- **CPU:** Minimum of a 64-bit processor with 2 cores. A processor with 4 or more cores is recommended for optimal performance.
- **RAM:** Minimum of 8 GB.

2.3. Software Dependencies

All dependencies are managed automatically by the Conda installation script.

- **Python:** Version 3.11 or newer.
- **GUI Framework:** PyQt5.
- **Cheminformatics:** RDKit.
- **Scientific Computing:** NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

3. Quick Start Guide

This section provides instructions for rapid installation and initial launch of PyRMD Studio.

3.1. Installation Process for Windows users

The installation script automates dependency management and system configuration.

1. Download the complete PyRMD Studio package and extract it to a directory where you have write permissions.
2. Double click on the PyRMD_Studio_Setup.exe file and follow the installation procedure.
3. The setup file will detect if Conda is installed and install Miniconda if necessary. It will then create a dedicated Conda environment for PyRMD Studio. The process typically takes ~10 minutes.
4. When prompted, select the **user-specific installation**. This is the recommended option as it does not require administrator privileges.

3.2. Installation Process for Linux users

The installation script automates dependency management and system configuration.

5. Download the complete PyRMD Studio package and extract it to a directory where you have write permissions.
6. Open a terminal window and navigate to the extracted directory.
7. Ensure you are logged in as a regular user, not as root.
8. Execute the installation script by running the following command:

   ```
   bash install_pyrmd_studio.sh
   ```

9. Follow the on-screen prompts. The script will detect if Conda is installed and install Miniconda if necessary. It will then create a dedicated Conda environment for PyRMD Studio. The process typically takes ~10 minutes.
10. When prompted, select the **user-specific installation**. This is the recommended option as it does not require administrator privileges.

Upon completion, the script will provide a summary of the installation and create desktop shortcuts and application menu entries for launching the software.

3.3. First Launch and Verification

Launch PyRMD Studio using your preferred method (desktop shortcut, application menu, or terminal command – in this case, open a new terminal and type “pyrmd_studio”). The main interface will appear, displaying the PyRMD Studio logo and module selection buttons.

4. Using PyRMD: Ligand-Based Virtual Screening (LBVS)

This workflow is divided into two main phases: training a predictive model (Benchmarking) and using that model to screen a database (Screening).

4.1. Phase 1: Benchmarking a Model

In this phase, you will train and validate a predictive model.

1. Launch the PyRMD Studio application.
2. On the main screen, click **PyRMD** to open the Ligand-Based Virtual Screening module.
3. Ensure the **Benchmark** tab is selected.
4. Verify that **Expert mode** (top-left corner) is unchecked for a simplified interface.
5. Click **Browse** next to **Output Directory** and select a folder where the benchmarking results will be saved.
6. It is possible to provide your own data, a ChEMBL database, or a combination of the two. To provide your own training data:
   - Ensure the **Use ChEMBL Dataset** checkbox is unchecked.
   - Click **Browse** to select your "actives" file (in .csv format).
   - Click **Browse** to select your "inactives" file (in .csv format).
7. To use data from the ChEMBL database:
   - Check the **Use ChEMBL Dataset** box.
   - Enter the **ChEMBL ID** for your target receptor to automatically fetch the data or click **Browse** to load a previously downloaded ChEMBL .csv file.
8. (Optional) Click **Browse** to select a "decoys" file in .csv format.
9. Select the training parameters for **Active ε Cutoff** and **Inactive ε Cutoff**. You can hover your mouse over each parameter for general information or refer to the PyRMD papers (see Section 1.1) for a detailed explanation of these crucial hyperparameters.
10. Along with the Activity thresholds, the **Inhibition Thresholds** can be selected by the user if needed.
    - If the user wants to include data from fixed dose assays (e.g., expressed as % of inhibition) from a ChEMBL source, it is important to flag the 'INHIBITION THRESHOLD' button. Then, thresholds for the inhibition rate parameters can be modified.
11. Click **Update Configuration** to save your settings to the backend .ini file.
12. Click **Run Benchmark**. A dialog will appear, allowing you to select the number of CPU cores to use for the calculation.
13. After the process is completed, the Benchmark Results window will automatically open. Users can view the results or load them to directly launch the subsequent Screening phase. NOTE: it is possible to load an already performed benchmarking experiment through the **Open Results CSV** button.

4.2. Phase 2: Screening a Database

Once you have a validated model, you can use it to screen a large chemical database for potential hits.

1. In the PyRMD module, select the **Screening** tab.
2. In the **Screening module** sub-tab, enter the parameters that correspond to your best-performing model from the benchmarking phase.
3. Click **Browse** next to **Database to Screen** and select the molecular database file you wish to screen.
4. (OPTIONAL) Screened compounds can be filtered (e.g., using the Lipinski's rule of 5) by flagging **'FILTER'** and then specifying various parameters.
5. Click **Update Configuration** to save your settings.
6. Click **Run Screening** to start the virtual screening calculation.
7. Once the screening is complete, select the **Screening analysis module** sub-tab to view and analyze the results. You can visualize and subdivide the results into preferred thresholds/RMD scores for more critical results.

5. Using PyRMD2Dock: Structure-Based Virtual Screening (SBVS)

This protocol uses docking scores to train an RMD model, which can then be used to rapidly screen ultra-large databases.

5.1. Phase 1: Benchmarking a Model

1. Launch the PyRMD Studio application.
2. On the main screen, click **PyRMD2Dock** to open the Structure-Based Virtual Screening module.
3. Ensure the **Benchmark** tab is selected and that **Expert mode** is unchecked.
4. Click **Browse** next to **Output Directory** and select a folder for the results.
5. It is possible to use docking data already pre-processed by PyRMD, or to process and analyze unprepared docking results. To process raw docking results:
   - Ensure the **Use Single Docking Score File** box is selected.
   - Click **Browse** and upload the docking score file, which must contain three columns: a "Title" column which represents compound ID, one containing docking scores, the last containing the corresponding SMILES. Select the corresponding columns for each file, then click on **Merge** to generate a single, prepared file in .csv format.
   - Click **Analyze** to select the actives and inactives for generating the training sets.
6. To use pre-processed docking data:
   - Check the **Use Single Docking Score File** box (i.e., a single file containing the docking score column and the SMILES column along with the IDs).
   - Click **Browse** to upload a single .csv file containing SMILES strings ("Smiles" column), their IDs ("Title" column) and their corresponding docking scores.
   - Click **Analyze** to select active and inactive compounds for training.
7. Select thresholds for actives and inactives.
   - To select an **Active threshold** user can move the slider on the visible plot to select the compounds based upon their Docking Score. It is also possible to select the compounds by choosing their exact number or a percentage of the entire dataset.
   - **Inactive threshold** is selected in the same way.
8. Follow steps 8-13 from the **PyRMD Benchmarking** guide (Section 4.1) to configure parameters, run the benchmark, and analyze the results.

5.2. Phase 2: Screening a Database

1. In the PyRMD2Dock module, select the **Screening** tab.
2. Follow steps 2-7 from the **PyRMD Screening** guide (Section 4.2) to configure the screening with your best model, run the calculation, and analyze the results. NOTE: it is possible to load an already performed benchmarking experiment through the Open Results CSV button.

6. Step-by-Step Example: Simulating a LBVS Experiment with PyRMD

In this section we provide all the necessary files to simulate step-by-step a real LBVS experiment using PyRMD.

6.1. Phase 1: Benchmarking a Model

In this phase, you will train and validate a predictive model using a known set of compounds from a ChEMBL database.

1. Launch the PyRMD Studio application (or type 'pyrmd_studio' in a terminal).
2. On the main screen, click **PyRMD** to open the Ligand-Based Virtual Screening module.
3. Ensure the **Benchmark** tab is selected.
4. Verify that **Expert mode** (top-left corner) is unchecked for a simplified interface.
5. Click **Browse** next to **Output Directory** and select a folder where the benchmarking results will be saved.
6. To use data from the ChEMBL database:
   - Check the **Use ChEMBL Dataset** box is flagged.
   - Click **Browse** to load a previously downloaded ChEMBL .csv file.
     *Click → Browse → Tutorial → PyRMD → Benchmarking → fgfr1_chembl.csv
7. Click **Browse** to select a "decoys" file in .csv format.
   *Click → Browse → Tutorial → PyRMD → Benchmarking → fgfr1_decoys.csv
8. Select the training parameters for **Active Epsilon Cutoff** and **Inactive Epsilon Cutoff**.
   *For this tutorial, set Active Epsilon Cutoff to 0.95 and Inactive Epsilon Cutoff to 0.84
9. Flag **Inhibition Thresholds** and leave their default values.
10. You can hover your mouse over each parameter for general information or refer to the PyRMD paper (see Section 1.1) for a detailed explanation of these crucial hyperparameters
    *Refer to /Tutorial/PyRMD/Benchmarking/configs/0.95_0.84.ini for parameters we used for comparative results.
11. Click **Update Configuration** to save your settings to the backend .ini file.
12. Click **Run Benchmark**. A dialog will appear, allowing you to select the number of CPU cores to use for the calculation.
13. After the process is completed, the Benchmark Results window will automatically open. Users can view the results or load them to directly launch the subsequent Screening phase. NOTE: it is possible to load an already performed benchmarking experiment through the Open Results CSV button.
    *Refer to /Tutorial/PyRMD/Benchmarking/Benchmark_Results.csv to compare model training results.

6.2. Phase 2: Screening a Database

Once you have a validated model, you can use it to screen a large chemical database for potential hits.

1. After the benchmarking process is completed, the Benchmark Results window will automatically open. Click on **Run Screening** to directly perform screening with the selected model.
2. Otherwise, users can manually input the parameters. To do so, in the PyRMD module select the **Screening** tab and enter the parameters that correspond to your best-performing model from the benchmarking phase (check the Benchmark_Results.csv file).
3. Click **Browse** next to **Database to Screen** and select the molecular database file you wish to screen.
   *Click → Browse → Tutorial → PyRMD → Screening → database_to_screen.csv
4. For this experiment, we will not use any filter, so unflag **'FILTER'** box.
5. Click **Update Configuration** to save your settings.
6. Click **Run Screening** to start the virtual screening calculation.
7. Once the screening is complete, select the **Screening analysis module** sub-tab to view and analyze the results.
   *Click → Browse → Tutorial → PyRMD → Screening → Database_Predictions.csv

You can visualize the data and subdivide the results into preferred thresholds/RMD scores if preferred.

7. Best Practices: Managing Structural Homo/Heterogeneous Datasets

PyRMD Studio uses rigorous Butina clustering by default to split training and test sets. This prevents *data leakage* (where the model memorizes overlapping structural motifs rather than learning generalizable rules). However, highly homo/heterogeneous datasets require careful handling.

To ensure optimal performance, we highly recommend inspecting the clustering logs generated after the Benchmarking phase to assess your dataset's structural topology.

Based on the logs, you might encounter two extreme topological scenarios that negatively affect model training:

7.1. The "Dominant Scaffold" (High Homogeneity):

- *Symptom:* A single cluster contains >50-60% of your active compounds.
- *Impact:* The Butina algorithm will put this entire massive cluster in either the train or test set. The model will likely suffer a significant performance drop because it is forced to cross a massive structural gap without overlapping information.
- *Action:* Your dataset lacks diversity. You can use **Expert Mode** to manually **increase** the Butina similarity threshold (e.g., from 0.65 to 0.80). This stricter criterion forces the algorithm to break the excessively large cluster into smaller, more distinct sub-scaffolds. Alternatively, consider acquiring a more structurally diverse initial dataset.

7.2. The “Extreme Fragmentation” (High Heterogeneity):

- *Symptom:* The vast majority of your active compounds are classified as "singletons" (clusters of size 1).
- *Impact:* The dataset is too structurally noisy. The model may struggle to find any common structural denominator to learn from.
- *Action:* Use **Expert Mode** to manually **decrease** the similarity threshold (e.g., from 0.65 to 0.50) to group looser neighbors together into functional clusters, or stringently pre-filter your dataset to remove false positives and promiscuous binders.

8. Troubleshooting

This section addresses common issues you may encounter.

Symptom	Possible Cause(s)	Solution(s)
Installation script fails	Missing system dependencies or insufficient permissions.	1. Ensure you are running the script as a regular user, not root. 2. Verify you have write permissions in the target directory. 3. Check your internet connection.
Application does not launch	Conda environment is not activated correctly or there is a graphics driver issue.	1. Try launching the application from the terminal by typing “pyrmd_studio” and check for error messages. 2. Ensure your system's graphics drivers are up to date.
ChEMBL data fails to download	Network connectivity issue or incorrect ChEMBL ID.	1. Check your internet connection. 2. Double-check that the ChEMBL ID is correct.
Calculation runs very slowly	Insufficient hardware resources allocated.	1. When starting a calculation, ensure you allocate an appropriate number of CPU cores for your system. 2. Close other resource-intensive applications.